YILU WANG

M.Sc. Candidate, Molecular Biosciences (Cancer Biology) · DKFZ & Heidelberg University
yiluwang2002@gmail.com · +49 175 743 9605 · yiluw.github.io · Heidelberg, Germany

EDUCATION

Oct 2025 – Present **Heidelberg University & German Cancer Research Center (DKFZ)**
M.Sc. in Molecular Biosciences · Major: Cancer Biology
- Grade average 1.4 / 1.0 (top band "very good"; 1.0 = highest).
- Hans-Peter Wild Talent Scholarship (2025).

Sep 2021 – Jun 2025 **Tsinghua University, Beijing, China**
B.S. in Biochemistry & B.E. in Biomedical Engineering (double major)
- Grade average 3.7 / 4.0 (top 10%).
- University excellence scholarships (2022 – 2024).

RESEARCH EXPERIENCE

Jun 2026 – Present **Institute for Computational Biomedicine & TSPC, Heidelberg University Hospital**
Research Intern · Advisor: Prof. Denis Schapiro (Spatial omics)
- Comparison and optimization of data-integration methods for imaging-based spatial proteomics.

Nov 2024 – Feb 2025 **Harvard University — School of Engineering & Applied Sciences**
Research Intern · Advisor: Prof. David Weitz (Experimental soft condensed matter)
- Developed hydrogel-bead–based in vitro display systems for nanobody discovery; constructed and screened a ~10^{13}-variant library by high-throughput FACS, recovering binders with ultra-high affinity and specificity.

Jul – Aug 2024 **Stanford University — Department of Mechanical Engineering**
Research Intern · Advisor: Prof. Juan Santiago (Microfluidics)
- Built tube-embedded microfluidic devices for electrokinetic cell-free DNA purification (isotachophoresis) with TaqMan qPCR readout; imaged single-molecule DNA dynamics in PDMS chips.

Sep 2022 – Jun 2024 **Tsinghua University — Institute of Medical Systems Biology**
Undergraduate Researcher · Advisor: Prof. Yong Guo (Molecular diagnostics)
- Developed multi-target digital PCR assays for robust HER2 detection in FFPE and blood samples, with a Python watershed pipeline for droplet clustering.

PUBLICATIONS

[1] Q. Jiang, X. Zang, **Y. Wang**, A. S. Avaro, D. A. Huyke, J. G. Santiago. *"A three-dimensional microfluidic device embedded within a thermal cycler tube for electrokinetic DNA extraction."* Lab on a Chip, 2025.

[2] **Y. Wang**, Y. Guo. *"Single-tube single-, dual-, and triple-target digital PCR assays for detection of HER2 amplification."* B.S. Thesis, Tsinghua University, 2025.

[3] X. Chen, et al., **Y. Wang**, Y. Guo, Z. Xie. *"Development and evaluation of a multiplex digital PCR method for sensitive and accurate detection of respiratory pathogens in children."* Virology, 2024.

[4] H. Yang, et al., **Y. Wang**, G. Huang. *"Graphene-based flexible biosensing technology and wearable precision medical-health-monitoring application."* Chinese Journal of Lasers, 2024.

SCHOLARSHIPS & AWARDS

2025	Hans-Peter Wild Talent Scholarship, Heidelberg University
2022 – 2024	Comprehensive Excellence Scholarship, Tsinghua University (top 2% of undergraduates)
2023	Academic Research Excellence Scholarship, Tsinghua University (top 1% of undergraduates)
2023	Gold Medal & nominations for Best Therapeutics Project / Best Software Tool, iGEM Competition
2024	Second Prize, Tsinghua 42nd Challenge Cup (Student Academic & Scientific Works)
2023	Independent Investigator, Qiyan Program Fund, Beijing Natural Science Foundation
2020	National high-school Olympiads: First Prize in Math & Physics (Heilongjiang); Silver Medal, China Girls' Mathematics Olympiad

EXTRACURRICULAR ACTIVITIES

2022 – 2025

Qingxin Times, Tsinghua University

Chief Reporter, Director & Consultant

- Named Most Influential Student Journalist (2022); awarded University Journalism Awards for Best Writing (2023); wrote, edited and published 20+ feature articles across campus and external press.

2023

Rural Revitalization Workstation, Tsinghua University

Team Leader

- Led two field teams researching and supporting rural communities in Shaanxi, China; awarded Best Student Social Practice (2022, 2023).

2023 – 2024

International Genetically Engineered Machine (iGEM)

Team Leader (2023) & Advisor (2024)

- Led and mentored synthetic-biology therapeutics teams.

GRADUATE LABORATORY TRAINING

Winter 2025

Cancer immunology, stem-cell biology & functional genomics

Practical course (HP-F2), DKFZ

- Three lab rotations: engineering and cytotoxicity testing of CAR-T cells (P. Schmidt); inflammatory-stress responses of blood stem cells (M. Essers); and RNA-seq differential-expression analysis in R (C. Erkut).

Summer 2026

Cell reprogramming, translation & RNA modification

Practical course (HP-F7), DKFZ

- Three lab rotations: reprogramming stem cells into neurons with chromatin (ATAC-seq) profiling (M. Mall & D. Odom); ribosome profiling of protein translation (F. Loayza-Puch); and m^6A RNA-methylation effects in melanoma (M. Frye).

PROFESSIONAL SKILLS

Experimental	Molecular & cell biology (PCR / qPCR / digital PCR, cloning, electrophoresis, western blot, ELISA); cell culture (cancer lines, T cells, iPSCs, neurons); flow cytometry / FACS; microscopy (confocal, immunofluorescence); microfluidics (PDMS design & fabrication, droplet sorting).
Computational	NGS analysis — RNA-seq, ATAC-seq, Ribo-seq, gene-enrichment analysis; machine learning / neural networks. Programming: Python, R, C/C++, MATLAB.
Languages	Chinese (native) · English (fluent) · Japanese (intermediate) · German (elementary)